

# A disproof of Conjecture 00000001934

$\mathrm{SL}(n, \mathbb{Z})$  acts analytically on  $S^{n-1}$  with infinite image

## 1 The conjecture

Conjecture 00000001934 reads (English version; the Chinese version says the same):

*Definition:* The Zimmer program classifies actions of higher-rank lattices on low-dimensional manifolds. *Conjecture:* All actions of  $\mathrm{SL}(n, \mathbb{Z})$  ( $n \geq 3$ ) on  $(n-1)$ -manifolds are finite (the image is a finite group); the classification table of standard forms in  $n$  dimensions is exhausted by fractional-linear actions on projective space.

The conjecture is the conjunction of two clauses:

- (1) for every  $n \geq 3$ , every action of  $\mathrm{SL}(n, \mathbb{Z})$  on an  $(n-1)$ -dimensional manifold  $M$  has finite image, i.e. the homomorphism  $\mathrm{SL}(n, \mathbb{Z}) \rightarrow \mathrm{Perm}(M)$  defined by the action has finite range;
- (2) “the classification table of standard forms in  $n$  dimensions is exhausted by fractional-linear actions on projective space”.

Clause (2) is not a precise statement and we do not need it: the conjunction is false as soon as clause (1) is false. Clause (1) is an unconditional assertion (“All actions”), not an example or a parenthetical; the parenthesis only explains what “finite” means. Clause (1) contains no volume-preservation hypothesis.

**Reading.** We refute clause (1) under the following reading, uniformly in the regularity. Fix  $r \in \{0, 1, 2, \dots, \infty, \omega\}$ . “Action on an  $(n-1)$ -manifold” means a group action of  $\mathrm{SL}(n, \mathbb{Z})$  on a  $C^r$  manifold  $M$  (without boundary) of dimension  $n-1$  in which every group element acts by a  $C^r$  map ( $r=0$ : a continuous action on a topological manifold;  $r=\infty$ : smooth;  $r=\omega$ : real-analytic). We refute the clause even when  $M$  is in addition required to be compact, connected and Hausdorff (these extra requirements only weaken the clause). Our counterexample works for *every*  $n \geq 3$  and every  $r$  at once, because it is real-analytic on a closed connected manifold.

## 2 The counterexample

Let  $n \geq 2$  and let  $S^{n-1} = \{v \in \mathbb{R}^n : \|v\| = 1\}$  be the unit sphere (Euclidean norm). It is a compact real-analytic manifold of dimension  $n-1$  (stereographic charts), connected when  $n \geq 3$ . For  $A \in \mathrm{SL}(n, \mathbb{Z})$  let  $L_A : \mathbb{R}^n \rightarrow \mathbb{R}^n$  be  $v \mapsto Av$  (entries cast to  $\mathbb{R}$ ) and define

$$A \cdot v = \frac{Av}{\|Av\|} \quad (v \in S^{n-1}).$$

**Lemma 1.** *This is a well-defined group action of  $\mathrm{SL}(n, \mathbb{Z})$  on  $S^{n-1}$ .*

*Proof.*  $L_A$  is injective since  $L_{A^{-1}}L_A = \text{id}$ , so  $Av \neq 0$  for  $v \neq 0$  and  $A \cdot v \in S^{n-1}$ . For  $v \in S^{n-1}$ ,  $I \cdot v = v/\|v\| = v$ . For the composition law, put  $c = \|Bv\|^{-1} > 0$ ; then  $A \cdot (B \cdot v)$  is the radial projection of  $L_A(cBv) = cABv$ , and radial projection is invariant under positive scaling:  $\|cw\|^{-1}cw = \|w\|^{-1}w$ . Hence  $A \cdot (B \cdot v) = (AB) \cdot v$ .  $\square$

**Lemma 2.** *Each  $A \in \text{SL}(n, \mathbb{Z})$  acts by a real-analytic ( $C^\omega$ ) map  $S^{n-1} \rightarrow S^{n-1}$ .*

*Proof.* The map  $g(w) = \|L_A w\|^{-1}L_A w$  is analytic on  $\{w : L_A w \neq 0\} \supset S^{n-1}$ , because  $L_A$  is linear, the Euclidean norm is analytic away from 0, and inversion and scalar multiplication are analytic. The inclusion  $S^{n-1} \hookrightarrow \mathbb{R}^n$  is analytic, and a  $C^\omega$  map into  $\mathbb{R}^n$  with values in  $S^{n-1}$  is  $C^\omega$  as a map into the manifold  $S^{n-1}$ .  $\square$

**Lemma 3.** *Let  $n \geq 2$  and  $i \neq j$ . For  $k \in \mathbb{Z}$  let  $T_k = I + kE_{ij} \in \text{SL}(n, \mathbb{Z})$  (a transvection,  $\det T_k = 1$ ). Then the permutations  $v \mapsto T_k \cdot v$  of  $S^{n-1}$  are pairwise distinct; hence the image of  $\text{SL}(n, \mathbb{Z})$  in  $\text{Perm}(S^{n-1})$  is infinite.*

*Proof.*  $T_k e_j = k e_i + e_j$ , so  $T_k \cdot e_j = c_k(k e_i + e_j)$  with  $c_k = \|T_k e_j\|^{-1} > 0$ . If  $T_k \cdot e_j = T_l \cdot e_j$ , the  $j$ -th coordinates give  $c_k = c_l$  and then the  $i$ -th coordinates give  $c_l k = c_l l$ , so  $k = l$ . Thus  $k \mapsto (v \mapsto T_k \cdot v)$  is injective on  $\mathbb{Z}$ .  $\square$

**Theorem 4.** *For every  $n \geq 3$  and every regularity  $r$ , clause (1) fails:  $S^{n-1}$  is a compact connected Hausdorff  $C^r$  manifold of dimension  $n-1$ ,  $\text{SL}(n, \mathbb{Z})$  acts on it by  $C^r$  maps (Lemmas 1, 2), and the image of the action is infinite (Lemma 3). Consequently the conjecture (the conjunction of clauses (1) and (2)) is false, whatever clause (2) means.*

*Remark 5.* The counterexample is the double cover of the projective action of  $\text{SL}(n, \mathbb{Z})$  on  $\mathbb{R}P^{n-1}$ . It is the standard reason why the dimension bound in Zimmer’s conjecture is  $\dim M < n-1$ , not  $\leq n-1$ . For instance, Brown–Fisher–Hurtado (arXiv:1608.04995, abstract) prove, for a cocompact lattice  $\Gamma$  in  $\text{SL}(n, \mathbb{R})$ , that “any homomorphism  $\rho: \Gamma \rightarrow \text{Diff}(M)$  has finite image if the dimension of  $M$  is less than  $n-1$  and that any homomorphism  $\rho: \Gamma \rightarrow \text{Diff}(M, \omega)$  has finite image if the dimension of  $M$  is less than  $n$ ”. Thus a volume-preservation hypothesis would change the problem; the conjecture has none. The witness is an action on the sphere, not on projective space, so it is not one of the “fractional-linear actions on projective space” mentioned in clause (2).

### 3 Lean formalization

File `lean/Conjecture1934/Basic.lean` (Lean 4, Mathlib, only `import Mathlib`; namespace `C1934`). The sphere is Mathlib’s `Metric.sphere (0 : EuclideanSpace R (Fin (d+1))) 1` with Mathlib’s analytic manifold structure modelled on `EuclideanSpace R (Fin d)` (`Geometry/Manifold/Instances/Sphere.lean`);  $\text{SL}(n, \mathbb{Z})$  is `Matrix.SpecialLinearGroup (Fin n) Z`.

- `lin, nrm, act`:  $L_A$ , radial projection and  $A \cdot v$ .
- `instMulAction`: Lemma 1 (a `MulAction` instance).
- `contMDiff_smul`: Lemma 2, `ContMDiff (R d) (R d) omega (fun x => A . x)`.
- `tv, tv_smul_injective, range_toPermHom_infinite`: Lemma 3; the image is `Set.range (MulAction.toPermHom (SLZ n) (Sph n))`.
- `connectedSpace_sph`:  $S^d$  is connected for  $d \geq 1$  (compactness and the Hausdorff property are Mathlib instances).

- `Clause1At r n`: for all  $d$  with  $d+1 = n$  and all compact, connected, Hausdorff  $M : \text{Type}$  with `ChartedSpace (EuclideanSpace R (Fin d)) M`, `IsManifold (R d) r M` and a `MulAction (SLZ n) M` in which every element acts by a `ContMDiff` map of order  $r$ , the range of `MulAction.toPermHom` is finite. `Clause1 r` is `Clause1At r n` for all  $n \geq 3$ .
- Main theorems: `not_clause1At (r) (n) (hn : 3 <= n) : not (Clause1At r n)`;  
`not_clause1 (r) : not (Clause1 r)`; `not_conjecture (r) (Clause2 : Prop) : not (Clause1 r and Clause2)`.

Here  $r$  ranges over `WithTop ENat`: 0 is continuity, finite values are  $C^k$ , the value `(top : ENat)` is  $C^\infty$  and `omega` (the top element) is analyticity.

**Axiom audit.** `#print axioms` for `C1934.not_clause1At`, `C1934.not_clause1` and `C1934.not_conjecture` reports exactly `[propext, Classical.choice, Quot.sound]`. There is no `sorry` and no `native_decide`. The file builds with `lake build` on Lean v4.33.1 and the matching Mathlib.